Radian Measure and Arc Length

Section 1: Degrees and Radians Conversion

1 Convert 150° to exact radians in simplest form.

a. $\frac{5\pi}{6}$

b. $\frac{2\pi}{3}$

c. $\frac{3\pi}{4}$

d. $\frac{7\pi}{6}$

2 Convert $\frac{7\pi}{4}$ radians to degrees.

a. 315°

b. 225°

c. 300°

d. 330°

3 Find a positive angle coterminal with $-\frac{2\pi}{3}$ in the interval $[0, 2\pi)$.

a. $\frac{4\pi}{3}$

b. $\frac{\pi}{3}$

c. $\frac{5\pi}{3}$

d. $\frac{2\pi}{3}$

Section 2: Arc Length

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4 State the formula for arc length s subtended by a central angle θ (in radians) in a circle of radius r.

a. $s = r\theta$

b. $s = \frac{1}{2}r^2\theta$

c. $s = 2\pi r\theta$

d. $s = \frac{r}{\theta}$

5 Find the arc length intercepted by a central angle of $\frac{2\pi}{3}$ in a circle of radius 9 cm.

a. 6π cm

b. 3π cm

c. 18π cm

d. 12π cm

6 An arc of length 10 m subtends a central angle of 2 radians. What is the radius of the circle?

a. 5 m

b. 20 m

c. 10 m

d. 2.5 m

Section 3: Sector Area

7 State the formula for the area A of a sector of a circle with radius r and central angle θ (in radians).

a. $A = \frac{1}{2}r^2\theta$

b. $A = r^2\theta$

c. $A = \frac{1}{2}r\theta$

d. $A = \pi r^2\theta$

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8 Find the area of a sector with radius 6 in and central angle $\frac{\pi}{4}$.

a. $\frac{9\pi}{2}$ sq in

b. 9π sq in

c. $\frac{9\pi}{4}$ sq in

d. 18π sq in

9 If a sector of radius 4 cm has an area of 8π cm^2, find its central angle θ.

a. π radians

b. 2π radians

c. $\pi/2$ radians

d. 4π radians

Section 4: Linear and Angular Speed

10 What is the relationship between linear speed v, radius r, and angular speed ω (in rad/time)?

a. $v = r\omega$

b. $v = \frac{\omega}{r}$

c. $v = r^2\omega$

d. $v = 2\pi r\omega$

11 A wheel of radius 2 ft rotates at 60 rpm. Find its angular speed in radians per second.

a. 2π rad/s

b. 4π rad/s

c. π rad/s

d. 120π rad/s

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12 In the previous question ($r = 2$ ft, $\omega = 2\pi$ rad/s), what is the linear speed of a point on the rim?

a. 4π ft/s

b. 2π ft/s

c. 8π ft/s

d. 120 ft/s

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